

Abhineet Saha

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EDUCATION

- **Vellore Institute of Technology, AP** 2022 – 2026
• *B.Tech, Computer Science and Engineering (CGPA: 9.25)*

PROFESSIONAL EXPERIENCE

- **Cooper Square Technologies Inc. (Profound)**  Remote, India
Software Engineer – Independent Contractor Jan 2026 – Aug 2026
 - Designed and shipped a **route-binding migration engine** that transformed destructive URL changes into a confidence-scored, reviewable workflow, preventing broken content bindings.
 - Optimized **PostgreSQL/tRPC APIs** by eliminating N+1 queries, adding indexes and batching, and fixing cross-tenant scans across endpoints with **1.8–6.5s p99 latency**.
 - Improved platform **reliability and security** by authoring **34% of the test suite**, implementing multi-tenant database scoping, and remediating **2 SSRF vulnerabilities, 1 XSS vector, and 1 CVE**.
 - Rebuilt the **Media Library** using a blue-green migration and co-led the **Sanity-to-ProfoundCMS migration**, contributing approximately **48% of the CMS integration layer**.

PROJECTS

- **pyxTrace** 
 - *Python Runtime Tracing & Visualization Toolkit – Python, Typer, Streamlit, GitHub Actions*
 - Built and published an open-source Python profiler to **PyPI**, combining bytecode tracing, heap profiling, and OS syscall monitoring with real-time visualization.
 - Designed a **cross-platform tracing architecture** with pluggable Linux, macOS, Windows, and fallback backends for graceful platform compatibility.
 - Implemented configurable tracing modes to **balance profiling depth and runtime overhead**, enabling practical tracing of long-running workloads.
 - Automated **PyPI releases with GitHub Actions** and added JSONL session recording with offline replay for post-run analysis.
- **SafeRoute India – Accident-Aware Route Scoring Platform** 
 - *FastAPI, PostGIS, LightGBM, Redis, React, Docker*
 - Architected an **accident-aware route scoring platform** using Indian government accident datasets, scoring routes on a **0–100 safety scale** based on severity, distance, recency, and time-of-day risk.
 - Built a **PostGIS geospatial scoring engine** to analyze accident exposure along route corridors and identify safety hotspots across road networks.
 - Developed a **LightGBM risk prediction pipeline** using 24 engineered features covering accident severity, causal factors, temporal patterns, weather, and emergency infrastructure.
 - Delivered **13 REST APIs** for route scoring, hotspot analysis, departure-time optimization, risk forecasting, SOS context, and community incident reporting.

PROFESSIONAL CERTIFICATIONS

- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate:** Credential Verification
- **Microsoft Certified: Azure AI Engineer Associate:** Credential Verification

CO-CURRICULAR ACTIVITIES

- **Technical Lead – Mozilla Open Source Community, VIT-AP:** Led a cross-functional team of **12+ members**, contributing to global open-source projects and mentoring juniors on GitHub workflows and software architecture.
- **Finalist – Google Dev Sprint '25:** Built and presented a scalable prototype under time constraints, demonstrating technical execution, creativity, and design-led problem solving to industry leaders.

TECHNICAL SKILLS

- **Languages:** TypeScript, Python, JavaScript, Java, SQL
- **Backend:** Node.js, Express.js, FastAPI, tRPC
- **Frontend:** React, Next.js, Tailwind CSS
- **Databases:** PostgreSQL, PostGIS, Redis, Supabase
- **Tools:** Git, GitHub Actions, Docker, Linux, Bun